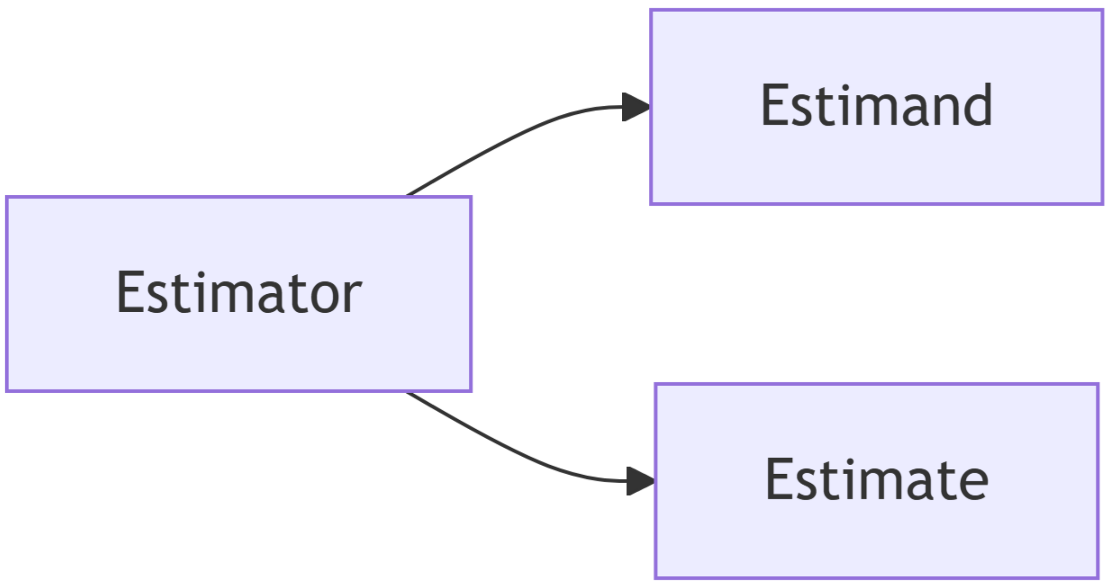
Properties of OLS and Statistical Inference

Lecture 3

In this lecture we investigate the properties of the OLS estimator and how it relates to the assumptions we make about the unobserved population regression model. As we do so it will help to keep in mind the distinction between the estimand, the estimator, and the estimate.



The estimand (β) is an unknown population parameter (i.e., a non-random number); the estimator ($\hat{\beta}$) a random variable; and the estimate (e.g. 0.187) a realized value of said random function (i.e. a non-random number).

Accuracy vs Effiency

Accuracy

Is $\hat{\beta} = \beta$?

- No!
- In fact, the statement does not make sense since $\hat{\beta}$ is a random variable while β is a non-random constant.
- If we are accurate, we can ask: $Pr(\hat{\beta} = \beta) = ?$
- The answer is: $Pr(\hat{\beta} = \beta|\mathbf{x}) = 0$ for any u_i with a continuous distribution (conditional on $\mathbf{x}$).

Can we **expect** – *a priori* – for $\hat{\beta}$ to be "on target"?

- This is a question of *biasedness*: is $E[\hat{\beta}] = \beta$?
- The answer: it depends!
- Specifically, it depends on the properties of the model!

> **Note**
>
> It is an all to common mistake to conflate the estimator - in particular, the OLS estimator - with the linear regression model. People will say, "We estimate an OLS regression model". This doesn't really make sense, since you OLS is the estimator, not the name of the model. You can say, "We estimate a linear regression model using OLS".
> This may seem pedantic, but it is an important distinction since there are a range of estimators that can be used to estimate a linear regression model; all with different properties. In addition, the OLS estimator can be applied to any data, regardless of the true model. So it is a good idea to make sure your language is accurate.

Biasedness

When is the OLS estimator unbias?

$$E[\hat{\beta}] = \beta$$

Under the assumptions MLR 1-4,

$$E[\hat{\beta}_j] = \beta_j \qquad j = 0, \dots, k$$

Proof

We do not have the math to do this for the multivariate case. However, we can show this for the univariate case:

Under MLR 1-4 (equiv. to Wooldridge's SLR 1-4)

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

where $E[u_i|x_i] = 0$.

The OLS estimator for β_1 is given by (see Lecture 2):

$$\begin{aligned}
\hat{\beta}_1 &= \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} \\
&= \frac{\sum_{i=1}^n y_i (x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} \\
&= \frac{\sum_{i=1}^n y_i (x_i - \bar{x})}{\text{SST}_x}
\end{aligned}$$

where $\text{SST}_x = \sum_{i=1}^n (x_i - \bar{x})^2$

Substitute the model ($y_i = \beta_0 + \beta_1 x_i + u_i$) in:

$$\begin{aligned}
\hat{\beta}_1 &= \frac{\sum_{i=1}^n (\beta_0 + \beta_1 x_i + u_i)(x_i - \bar{x})}{\text{SST}_x} \\
&= \frac{\beta_0}{\text{SST}_x} \sum_{i=1}^n (x_i - \bar{x}) + \frac{\beta_1}{\text{SST}_x} \sum_{i=1}^n (x_i - \bar{x}) x_i + \frac{1}{\text{SST}_x} \sum_{i=1}^n (x_i - \bar{x}) u_i
\end{aligned}$$

Now we use the fact that:

$$\hat{\beta}_1 = \frac{\beta_0}{\text{SST}_x} \underbrace{\sum_{i=1}^{n}(x_i - \bar{x})}_{=0} + \frac{\beta_1}{\text{SST}_x} \underbrace{\sum_{i=1}^{n}(x_i - \bar{x})x_i}_{\text{SST}_x} + \frac{1}{\text{SST}_x} \sum_{i=1}^{n}(x_i - \bar{x})u_i$$

To get,

$$\hat{\beta}_1 = \beta_1 + \frac{1}{\text{SST}_x} \sum_{i=1}^{n}(x_i - \bar{x})u_i$$

Now we can take the *conditional* expectation of $\hat{\beta}_1$

$$\begin{aligned} E[\hat{\beta}_1|x] =& E\left[\beta_1 + \frac{1}{\text{SST}_x} \sum_{i=1}^{n}(x_i - \bar{x})u_i \middle| x\right] \\ =& \beta_1 + E\left[\frac{1}{\text{SST}_x} \sum_{i=1}^{n}(x_i - \bar{x})u_i \middle| x\right] \\ =& \beta_1 + \frac{1}{\text{SST}_x} \sum_{i=1}^{n}(x_i - \bar{x}) \underbrace{E[u_i|x]}_{=0} \\ =& \beta_1 \end{aligned}$$

Since, $E[\hat{\beta}_1|x] = \beta_1$, then by the Law of Iterated Expectations (see CEF Material)

$$E[\hat{\beta}_1] = E[E[\hat{\beta}_1|x]] = E[\beta_1] = \beta_1$$

Intuition

Unbiasedness means that *in expectation* the OLS estimator will give you the true parameter.

- IF MLR 1-4 hold

It also means that we estimated the model over-and-over-again, drawing data from the true DGP, on average we will get the true value.

- For example, if we ran a Monte Carlo simulation ($R = 1000$) of the model:

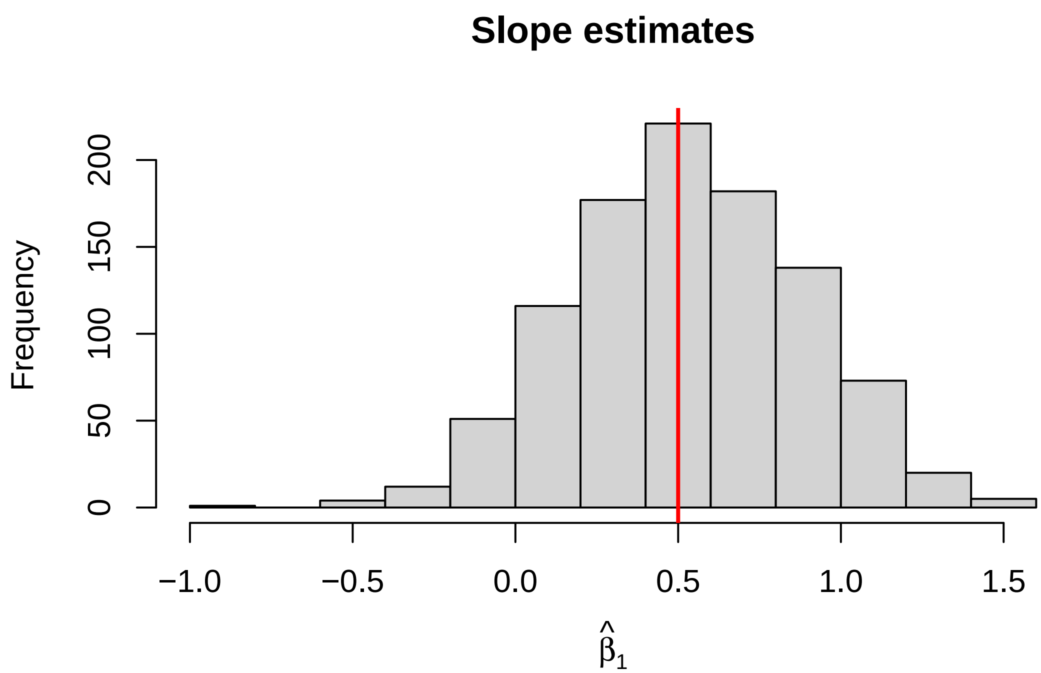
$$y_i = 2 + 0.5x_i + u_i$$

Monte Carlo simulation

```r
# Monte Carlo simulation of OLS estimator
# Model: y = 2 + 0.5*x + e,  e ~ N(0,1)

set.seed(123)      # for reproducibility
n <- 100           # sample size (set as desired)
R <- 1000    # number of Monte Carlo repetitions

b0 <- 2
b1 <- 0.5

# storage for estimates
b0_hat <- numeric(R)
b1_hat <- numeric(R)

# fixed regressor x (redrawn once, held fixed across repetitions)
x <- runif(n, min = 0, max = 1)

for (i in 1:R) {
  e <- rnorm(n, mean = 0, sd = 1)
  y <- b0 + b1 * x + e

  fit <- lm(y ~ x)

  b0_hat[i] <- coef(fit)[1]
  b1_hat[i] <- coef(fit)[2]
}

# Summary of results
#cat("Mean of intercept estimates:", mean(b0_hat), "\n")
cat("Mean of slope estimates:", mean(b1_hat), "\n")
```

```
Mean of slope estimates: 0.5282536
```

```
#cat("SD of intercept estimates:", sd(b0_hat), "\n")
#cat("SD of slope estimates:", sd(b1_hat), "\n")

# Plot sampling distributions
#par(mfrow = c(1, 2))
#hist(b0_hat, main = "Intercept estimates", xlab = expression(hat(beta)[0]))
#abline(v = b0, col = "red", lwd = 2)

hist(b1_hat, main = "Slope estimates", xlab = expression(hat(beta)[1]))
abline(v = b1, col = "red", lwd = 2)
```



Efficiency

Notice: there was a wide 'spread' of estimates in our simulation.

Why?

$$\hat{\beta}_1 \text{ is a random variable}$$

- which means it has a mean, variance, and distribution
- we have established that (under MLR 1-4) its mean is: $E[\hat{\beta}_1] = \beta_1$.

Variance

The variance of the estimator will depend on the variance of the x's and the error term. As before, we can show this more easily for the SLR:

$$
\begin{aligned}
Var(\hat{\beta}_1|x) =& E\Big[\big(\hat{\beta}_1 - \underbrace{E[\hat{\beta}_1|x]}_{\beta_1}\big)^2\Big|x\Big] \\
=& E[(\hat{\beta}_1 - \beta_1)^2|x]
\end{aligned}
$$

Earlier, we showed that

$$\hat{\beta}_1 = \beta_1 + \frac{1}{\mathrm{SST}_x}\sum_{i=1}^{n}(x_i - \bar{x})u_i$$

Substituting this is, we get:

$$Var(\hat{\beta}_1|x) = E\left[\left(\frac{1}{\mathrm{SST}_x}\sum_{i=1}^{n}(x_i - \bar{x})u_i\right)^2\middle|x\right]$$

After *some* math...(exploiting MLR-2)

$$Var(\hat{\beta}_1|x) = \frac{Var(u_i|x_i)}{SST_x} = \frac{\sigma^2}{SST_x}$$

Using MLR-5 (Homoskedasticity).

- The larger the variance of error term, the bigger the variance of the estimator
  - $\Rightarrow$ if you add variables that explain more of y you can reduce the variance.
- If you increase the variance in x, you reduce the variance of the estimator.

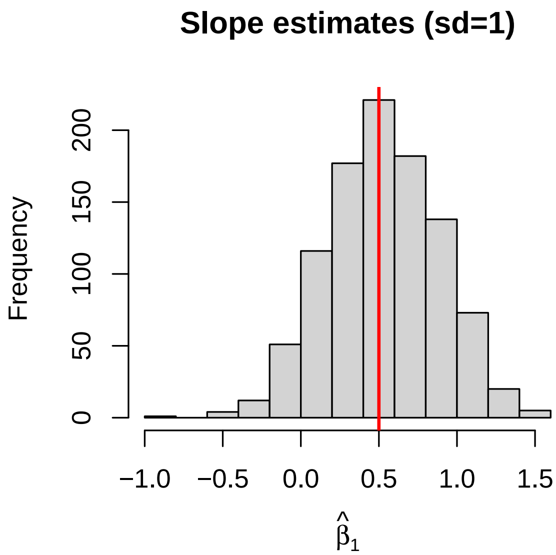
> Optimal experiment
>
> In Randomized Control Trials the regressor is a single dummy variable $w_i = 1$ $(= 0)$ if treated (if control). In this case, $SST_x = n\bar{w}(1 - \bar{w})$: a function that is quadratic in $\bar{w} = 1/n\sum_i w_i$ - the share of treated individuals. This function is concave, with a maximum at $\bar{w} = 0.5$. This tells us that the most efficient assignment is 50-50. With

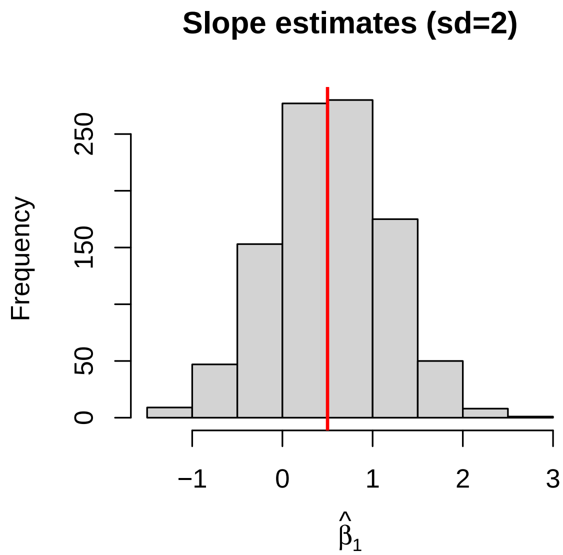
equal treatment, the variance of the estimator is minimized

Simulation

If we run the same Monte Carlo simulation for two cases - $\sigma = 1$ and $\sigma = 2$ - we should should see a big difference in the variance of the estimator.

```
# Monte Carlo simulation of OLS estimator
# Model: y = 2 + 0.5*x + e,  e ~ N(0,1)

# storage for estimates
b0_hat2 <- numeric(R)
b1_hat2 <- numeric(R)

# fixed regressor x (redrawn once, held fixed across repetitions)
x <- runif(n, min = 0, max = 1)

for (i in 1:R) {
  e <- rnorm(n, mean = 0, sd = 2)
  y <- b0 + b1 * x + e

  fit <- lm(y ~ x)

  b0_hat2[i] <- coef(fit)[1]
  b1_hat2[i] <- coef(fit)[2]
}

# Plot sampling distributions
par(mfrow = c(1, 2))
hist(b1_hat, main = "Slope estimates (sd=1)", xlab = expression(hat(beta)[1]))
abline(v = b1, col = "red", lwd = 2)

hist(b1_hat2, main = "Slope estimates (sd=2)", xlab = expression(hat(beta)[1]))
abline(v = b1, col = "red", lwd = 2)
```





Multivariate case

There is an equivalent formula for multivariate linear models:

$$Var(\hat{\beta}_j|\mathbf{x}) = \frac{Var(u_i|\mathbf{x})}{\text{SSR}_j} = \frac{\sigma^2}{\text{SSR}_j}$$

where SSR_j is the residual sum of squares from regressing x_j on all other regressors.

- if other regressors explain a lot of the variation in x_j, it $\downarrow \text{SSR}_j \Rightarrow \uparrow Var$
- adding regressors to a model has multiple effects in terms of efficiency.

Distribution

Recap

The OLS estimator is a **random variable**. Under MLR 1-5:

- $E[\hat{\beta}_j] = \beta_j \quad j = 0, \dots, k$
- $Var(\hat{\beta}_j) = \sigma^2 / SSR_j \quad j = 1, \dots, k$
- $Var(\hat{\beta}_0) =$

Normality

Under MLR 6:

$$\hat{\beta}_j|\mathbf{x} \sim \quad N(\beta_j, Var(\hat{\beta}_j|\mathbf{x}))$$

Why?

- $u_i|\mathbf{x} \sim N(0, \sigma^2)$
- conditional on x, $\hat{\beta}$ is a linear transformation of u.

Normalization

We can therefore standardize the distribution:

$$\frac{\hat{\beta}_j - \beta_j}{se(\hat{\beta}_j)}|x \sim N(0,1)$$

where $se(\hat{\beta}_j) = \sqrt{Var(\hat{\beta}_j)}$

This term - let's call it $z = (\hat{\beta}_j - \beta_j)/se(\hat{\beta}_j)$ is another random variable. Since we know the properties $N(0,1)$, we know:

- $Pr(z < \mathrm{z}_{0.025}) = 0.025$ where $\mathrm{z}_{0.025} = -1.96$ is the 2.5^{th} percentile of $N(0,1)$
- $Pr(z > \mathrm{z}_{0.975}) = 0.025$ where $\mathrm{z}_{0.975} = 1.96$ is the 97.5^{th} percentile of $N(0,1)$
- $Pr(\mathrm{z}_{0.025} < z < \mathrm{z}_{0.975}) = 0.95$ (or 95%)

Confidence Intervals

We can use this knowledge to form confidence intervals for the unknown β_j

$$\begin{aligned}
0.95 =& Pr(\mathrm{z}_{0.025} < z < \mathrm{z}_{0.975}) \\
=& Pr\left(\mathrm{z}_{0.025} < \frac{\hat{\beta}_j - \beta_j}{se(\hat{\beta}_j)} < \mathrm{z}_{0.975}\right) \\
=& Pr(\mathrm{z}_{0.025} \times se(\hat{\beta}_j) < \hat{\beta}_j - \beta_j < \mathrm{z}_{0.975} \times se(\hat{\beta}_j)) \\
=& Pr(-\hat{\beta}_j + \mathrm{z}_{0.025} \times se(\hat{\beta}_j) < -\beta_j < -\hat{\beta}_j + \mathrm{z}_{0.975} \times se(\hat{\beta}_j)) \\
=& Pr(\hat{\beta}_j - \mathrm{z}_{0.975} \times se(\hat{\beta}_j) < \beta_j < \hat{\beta}_j - \mathrm{z}_{0.025} \times se(\hat{\beta}_j))
\end{aligned}$$

Notice, the percentiles end up on opposite sides.

For symmetric distributions, like Normal, $\mathrm{z}_{0.975} = -\mathrm{z}_{0.025}$. This gives us the simplified Confidence Interval:

$$\mathrm{CI}_{0.95} = [\hat{\beta}_j - \mathrm{z}_{0.975} \times se(\hat{\beta}_j), \hat{\beta}_j + \mathrm{z}_{0.975} \times se(\hat{\beta}_j)]$$

Note

The CI is depends on the estimator $\hat{\beta}_j$. Since the estimator is random variable, the CI is also random. For this reason, we can ask: what is the probability that β_j lies within the CI.

$$Pr(\beta_j \in \mathrm{CI}_{0.95}) = 0.95$$

If replaced β_j with another value it the probability would change. This is because the distribution of $\hat{\beta}_j$ is centered around β_j.

Tip

Suppose I estimate $\hat{\beta}_j = 0.35$ (SE = 0.12) and construct the 95% confidence interval:

$$[0.35 - \mathrm{z}_{0.975} \times 0.12, 0.35 + \mathrm{z}_{0.975} \times 0.12]$$

What is the $Pr(\beta_j \in [0.35 - \mathrm{z}_{0.975} \times 0.12, 0.35 + \mathrm{z}_{0.975} \times 0.12])$? You might be tempted to say 95%, but (from a frequentist perspective) you would be wrong. The answer is either

0 or 1.
Why? The interval you have constructed is not random; it is just two numbers, a lower and upper bound. The question is equivalent to asking what is $Pr(5 \in [1,4])$? In this case the answer is evidently 0, but if the interval were $[2,6]$ the answer would be 1.

Inference

Null hypothesis

How does this work?

1. Establish a null hypothesis

$$H_0 : \beta_j = 0$$

Note

This is the most common null hypothesis as it essentially states that there is no relationship between y and x_j. In empirical research, we are often trying to establish whether there is a (causal) relationship bewteen two variables. In order to test this theory we state a null hypothesis that we will reject if the evidence supports our claim.
Why this backward approach? The answer two fold:

1. The estimator is a random variable, with infinite support. Consistent with $\beta_j = 0$, the estimator can take on any real value $\mathbb{R} : (-\infty, \infty)$. This means that every value the estimator could take on, is consistent with every unknown value of the true β_j.
2. We solve this problem by restricting probability of rejecting the null hypothesis when it is true (Type 1 error). That is, we design a test to *rule out* H_0, not rule it in.

If you want to establish that there is a relationship between, for example, smoking and lung cancer, you need to rule out the hypothesis that there is no relationship.

Alternative hypothesis

2. The null determines the alternative:

$$H_1 : \beta_j \neq 0$$

If we reject H_0 we conclude H_1. It is NOT accurate to say that we 'prove' H_1.

Test Statistic

3. Design a test statistic with a known distribution.

We know that under MLR 1-6,

$$\frac{\hat{\beta}_j - \beta_j}{se(\hat{\beta}_j)} | x \sim N(0,1)$$

Under $H_0 : \beta_j = 0$

$$\text{Test-stat} = \frac{\hat{\beta}_j - 0}{se(\hat{\beta}_j)} | x \underset{H_0}{\sim} N(0,1)$$

> **!** Important
>
> In this case, the test statics only has a normal distribution under H_0. If H_0 is not true, then the test statistic has different distribution.

Significance level

4. Pick a significance level (α).[1] This is the permitted probability of a type 1 error:

$$\alpha = Pr(\text{Reject } H_0 | H_0 \text{ is true})$$

In Economics, we typically use $\alpha = \{0.01, 0.05, 0.10\}$.

[1] Also referred to as the 'size' of the test.

Rejection rule

4. Determine a rejection rule consistent with a chosen α.

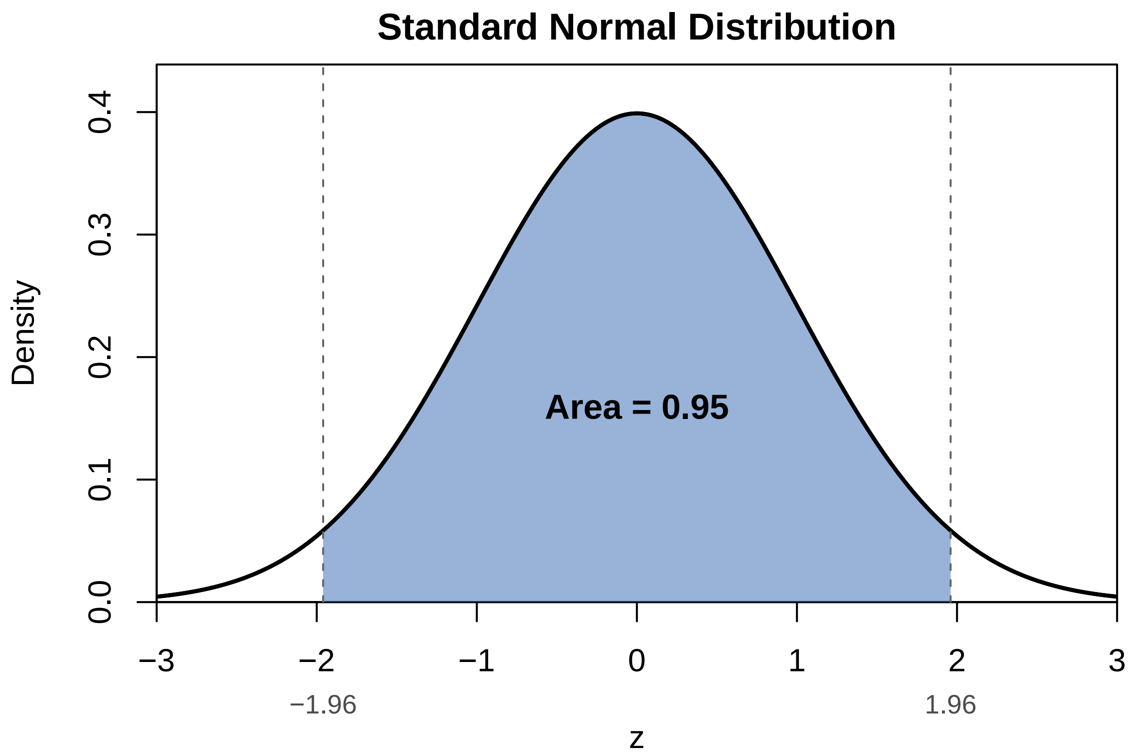
Under H_0,

$$\text{Test-stat} \underset{H_0}{\sim} N(0,1)$$

$$\Rightarrow Pr(-1.96 < \text{Test-stat} < 1.96|H_0) = 0.95$$

```
x <- seq(-3, 3, length.out = 1000)
y <- dnorm(x)

par(mar = c(4, 4, 2, 1))

plot(x, y, type = "l", lwd = 2, col = "black",
     xlim = c(-3, 3),
     xlab = "z", ylab = "Density",
     main = "Standard Normal Distribution",
     xaxs = "i", yaxs = "i", ylim = c(0, max(y) * 1.1))

# Shade the area between -1.96 and 1.96
x_shade <- seq(-1.96, 1.96, length.out = 500)
y_shade <- dnorm(x_shade)

polygon(c(x_shade[1], x_shade, tail(x_shade, 1)),
        c(0, y_shade, 0),
        col = rgb(0.20, 0.40, 0.70, 0.5), border = NA)

# Re-draw the curve on top so the outline stays crisp over the shading
lines(x, y, lwd = 2, col = "black")

# Reference lines at the cutoffs
abline(v = c(-1.96, 1.96), lty = 2, col = "gray40")

# Label the shaded area
text(0, dnorm(0) * 0.4, "Area = 0.95", cex = 1.1, font = 2)

# Label the cutoff values on the x-axis
```

```r
axis(1, at = c(-1.96, 1.96), labels = c("-1.96", "1.96"),
     line = 1.1, tick = FALSE, cex.axis = 0.85, col.axis = "gray30")
```

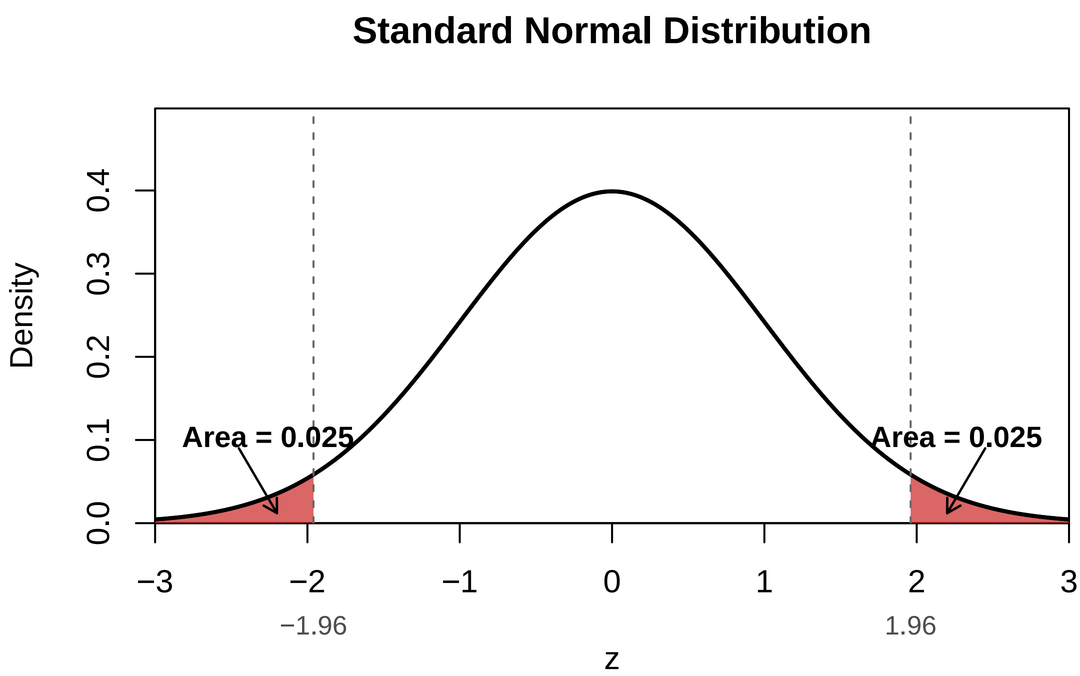


Reject H_0 if: Test-stat > 1.96 or Test-stat < -1.96

$$\Rightarrow \text{Reject if} \quad |\text{Test-stat}| > 1.96$$

```r
plot(x, y, type = "l", lwd = 2, col = "black",
     xlim = c(-3, 3),
     xlab = "z", ylab = "Density",
     main = "Standard Normal Distribution",
     xaxs = "i", yaxs = "i", ylim = c(0, max(y) * 1.25))

# --- Left tail: x <= -1.96 ---
x_left <- seq(-3, -1.96, length.out = 200)
y_left <- dnorm(x_left)
polygon(c(x_left[1], x_left, tail(x_left, 1)),
```

```
        c(0, y_left, 0),
        col = rgb(0.80, 0.15, 0.15, 0.7), border = NA)

# --- Right tail: x >= 1.96 ---
x_right <- seq(1.96, 3, length.out = 200)
y_right <- dnorm(x_right)
polygon(c(x_right[1], x_right, tail(x_right, 1)),
        c(0, y_right, 0),
        col = rgb(0.80, 0.15, 0.15, 0.7), border = NA)

# Re-draw the curve on top so the outline stays crisp over the shading
lines(x, y, lwd = 2, col = "black")

# Reference lines at the cutoffs
abline(v = c(-1.96, 1.96), lty = 2, col = "gray40")

# Labels for each tail, placed above the curve with an arrow pointing
# down into the shaded region (the tails are too thin to hold text)
arrows(x0 = -2.45, y0 = 0.09, x1 = -2.20, y1 = 0.012,
       length = 0.08, col = "black", lwd = 1.2)
text(-2.95, 0.10, "Area = 0.025", cex = 0.95, font = 2, pos = 4)

arrows(x0 = 2.45, y0 = 0.09, x1 = 2.20, y1 = 0.012,
       length = 0.08, col = "black", lwd = 1.2)
text(2.95, 0.10, "Area = 0.025", cex = 0.95, font = 2, pos = 2)

# Label the cutoff values on the x-axis
axis(1, at = c(-1.96, 1.96), labels = c("-1.96", "1.96"),
     line = 1.1, tick = FALSE, cex.axis = 0.85, col.axis = "gray30")
```



Type 2 errors

We've limited the probability of type 1 errors to α, but what about y**type 2 errors**?

$$Pr(\text{Fail to reject } H_0 | H_0 \text{ is false}) = ?$$

What do we mean by H_0 is false?

- not $H_0 \Rightarrow \beta_j \neq 0$; a lot of possibilities.

Power function

Suppose $\beta_j = \kappa$.

$$Pr(\text{Reject } H_0 | \beta_j = \kappa)$$

We call this the **power function**. And,

$$1 - Pr(\text{Reject } H_0 | \beta_j = \kappa) = Pr(\text{Type 2 error})$$

- Higher power $\rightarrow$ lower probability of type 2 error.

We know what this function looks like:

$$\begin{aligned}
Pr(\text{Reject } H_0|\beta_j=\kappa) =& Pr(|\text{Test-stat}|>1.96|\beta_j=\kappa)\\
=& Pr\left(\left|\frac{\hat{\beta}_j-0}{se(\hat{\beta}_j)}\right|>1.96\middle|\beta_j=\kappa\right)\\
=& Pr\left(\frac{\hat{\beta}_j-0}{se(\hat{\beta}_j)}<-1.96\middle|\beta_j=\kappa\right)+Pr\left(\frac{\hat{\beta}_j-0}{se(\hat{\beta}_j)}>1.96\middle|\beta_j=\kappa\right)
\end{aligned}$$

This probablity is not 0.05 because $\beta_j \neq 0$.

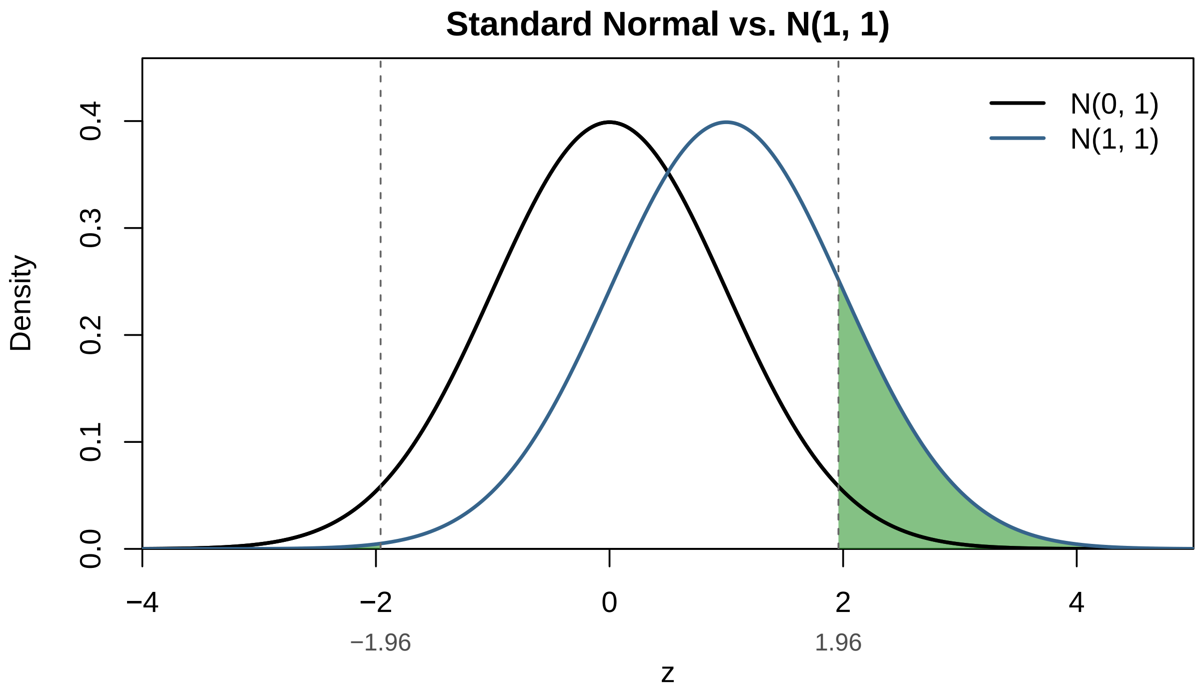
We have to add and subtract $\kappa/se(\hat{\beta}_j)$.

$$\begin{aligned}
Pr(\text{Reject } H_0|\beta_j=\kappa) =& Pr\left(\frac{\hat{\beta}_j-\kappa}{se(\hat{\beta}_j)}+\frac{\kappa-0}{se(\hat{\beta}_j)}<-1.96\middle|\beta_j=\kappa\right)+Pr\left(\frac{\hat{\beta}_j-\kappa}{se(\hat{\beta}_j)}+\frac{\kappa-0}{se(\hat{\beta}_j)}>1.96\middle|\beta_j=\kappa\right)\\
=& Pr\left(\underbrace{\frac{\hat{\beta}_j-\kappa}{se(\hat{\beta}_j)}}_{\sim N(0,1)}<-1.96-\frac{\kappa}{se(\hat{\beta}_j)}\middle|\beta_j=\kappa\right)+Pr\left(\underbrace{\frac{\hat{\beta}_j-\kappa}{se(\hat{\beta}_j)}}_{\sim N(0,1)}>1.96-\frac{\kappa}{se(\hat{\beta}_j)}\middle|\beta_j=\kappa\right)\\
=& \Phi(-1.96-\kappa/se(\hat{\beta}_j))+1-\Phi(1.96-\kappa/se(\hat{\beta}_j))\\
\geq& 0.05
\end{aligned}$$

Under the null hypothesis we assumed the distribution of $\hat{\beta}_j$ was centered on 0, when it is actually κ. This yields a probability of rejection greater than $0.05=\alpha$.

Suppose $\kappa/se(\hat{\beta}_j)=1$. Conditional on $\beta_j=\kappa$, the standardized test statistic is actually centered around 1.

```r
x <- seq(-4, 5, length.out = 1000)
y0 <- dnorm(x, mean = 0, sd = 1)   # null distribution
y1 <- dnorm(x, mean = 1, sd = 1)   # alternative distribution

par(mar = c(4, 4, 2, 1))

plot(x, y0, type = "l", lwd = 2, col = "black",
     xlim = c(-4, 5),
     xlab = "z", ylab = "Density",
     main = "Standard Normal vs. N(1, 1)",
     xaxs = "i", yaxs = "i", ylim = c(0, max(y0, y1) * 1.15))

# --- Shade N(1,1) area left of -1.96 ---
x_left <- seq(-4, -1.96, length.out = 200)
y_left <- dnorm(x_left, mean = 1, sd = 1)
polygon(c(x_left[1], x_left, tail(x_left, 1)),
        c(0, y_left, 0),
        col = rgb(0.20, 0.60, 0.20, 0.6), border = NA)

# --- Shade N(1,1) area right of 1.96 ---
x_right <- seq(1.96, 5, length.out = 200)
y_right <- dnorm(x_right, mean = 1, sd = 1)
polygon(c(x_right[1], x_right, tail(x_right, 1)),
        c(0, y_right, 0),
        col = rgb(0.20, 0.60, 0.20, 0.6), border = NA)

# Draw both curves on top of the shading
lines(x, y0, lwd = 2, col = "black")
lines(x, y1, lwd = 2, col = "steelblue4")

# Vertical reference lines at the rejection-rule cutoffs
abline(v = c(-1.96, 1.96), lty = 2, col = "gray40")

# Label the cutoff values on the x-axis
axis(1, at = c(-1.96, 1.96), labels = c("-1.96", "1.96"),
     line = 1.1, tick = FALSE, cex.axis = 0.85, col.axis = "gray30")

legend("topright", inset = 0.02,
       legend = c("N(0, 1)", "N(1, 1)"),
       col = c("black", "steelblue4"), lwd = 2, bty = "n")
```

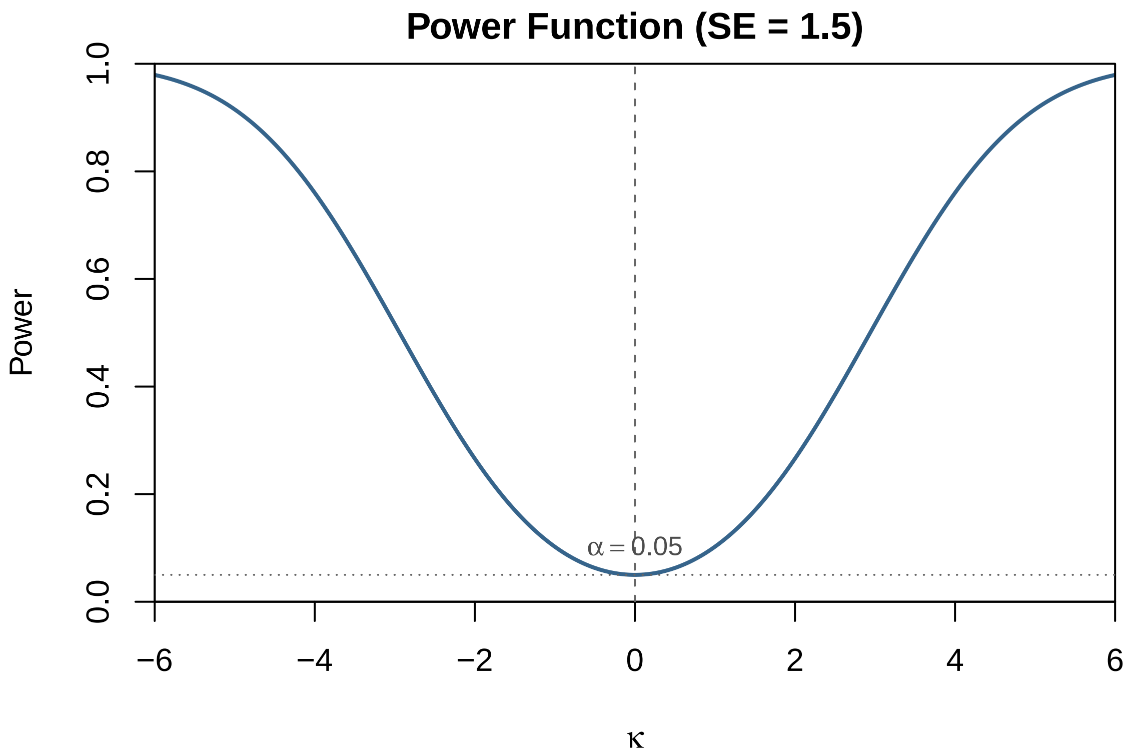


The probability of rejecting increases as κ moves further from 0 (the value under H_0).

Power function

```
power_fn <- function(kappa, se) {
  1 - pnorm(1.96 - kappa / se) + pnorm(-1.96 - kappa / se)
}

kappa <- seq(-6, 6, length.out = 1000)
power <- power_fn(kappa, se = 1.5)

par(mar = c(4, 4, 2, 1))

plot(kappa, power, type = "l", lwd = 2, col = "steelblue4",
     xlim = c(-6, 6), ylim = c(0, 1),
     xlab = expression(kappa), ylab = "Power",
     main = "Power Function (SE = 1.5)",
     xaxs = "i", yaxs = "i")

# Reference line at kappa = 0 (the null) and the significance level
abline(v = 0, lty = 2, col = "gray40")
```

```r
abline(h = 0.05, lty = 3, col = "gray40")

text(0, 0.05, expression(alpha == 0.05), pos = 3, cex = 0.85, col = "gray30")
```

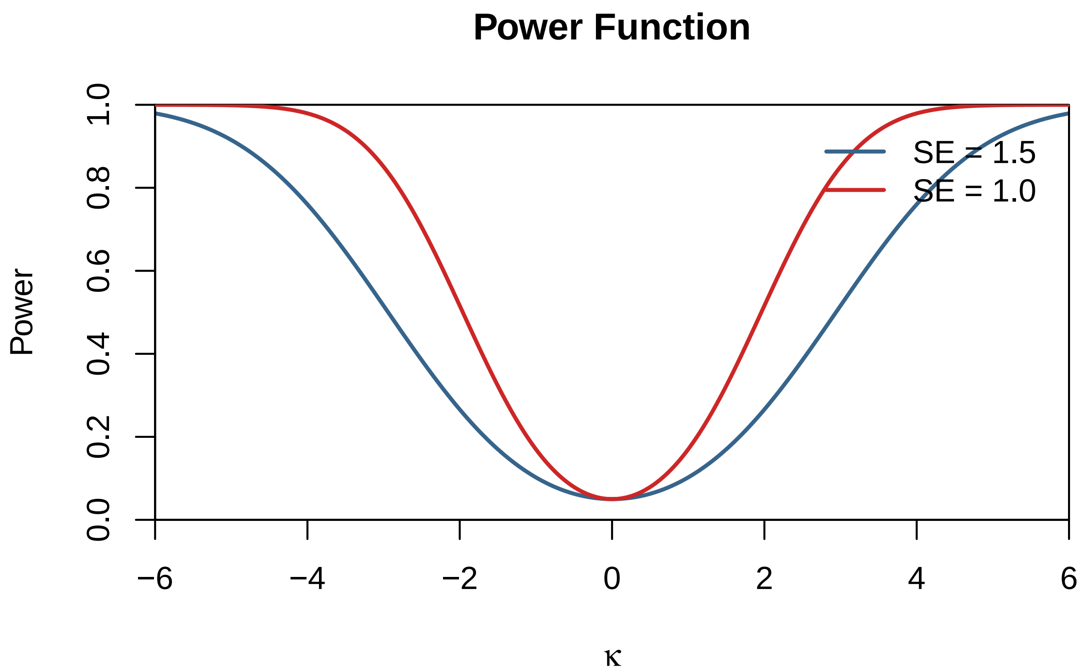


More power

One way to increase power is to reduce the SE.

```r
power_15 <- power_fn(kappa, se = 1.5)
power_10 <- power_fn(kappa, se = 1.0)

plot(kappa, power_15, type = "l", lwd = 2, col = "steelblue4",
     xlim = c(-6, 6), ylim = c(0, 1),
     xlab = expression(kappa), ylab = "Power",
     main = "Power Function", xaxs = "i", yaxs = "i")

lines(kappa, power_10, lwd = 2, col = "firebrick3")

legend("topright", inset = 0.02,
```

```
legend = c("SE = 1.5", "SE = 1.0"),
col = c("steelblue4", "firebrick3"), lwd = 2, bty = "n")
```



Variance estimator

'Sleight of hand'

All our computations have assumed knowledge of,

$$Var(\hat{\beta}_j|\mathbf{x}) = \frac{Var(u_i|\mathbf{x})}{\text{SSR}_j} = \frac{\sigma^2}{\text{SSR}_j}$$

BUT,

$$Var(u_i|\mathbf{x}) = \sigma^2$$

Is an *unknown* population parameter.

Estimator

We replace σ^2 with its estimator:

$$s^2 = \frac{1}{n-k-1}\sum_{i=1}^{n}\hat{u}_i^2$$

Why $n-k-1$?

- $\frac{1}{n}\sum_{i=1}\hat{u}_i^2$ is a **biased** estimator for σ^2
- $n-k-1$ are referred to as the residual *degrees of freedom*

t-distribution

When we substitue σ^2 for s^2, the recentered estimator has a (Student's) t-distribution

$$\frac{\hat{\beta}_j-\beta_j}{\widehat{se}(\hat{\beta}_j)}|x \sim T_{n-k-1}$$

where,

$$\widehat{se}(\hat{\beta}_j) = \sqrt{\frac{s^2}{SSR_j}}$$

> Student's t-distribution
>
> If $z \sim N(0,1)$ and $w \sim \chi^2_m$ then,
>
> $$\frac{z}{w} \sim T_m$$
>
> Note, a chi-squared distribution is defined as the sum of squared standard normals
>
> $$\sum_{l=1}^{m} z_l^2 \sim \chi^2_m$$

t-statistic

For $H_0 : \beta_j = 0$

$$\text{Test-stat} = \frac{\hat{\beta}_j}{\widehat{se}(\hat{\beta}_j)} | x \underset{H_0}{\sim} T_{n-k-1}$$

This is often referred to as the **t-statistic** in is reported as a default test parameter in regression output.

Example

Stata

```
* Open data
use "data/lcf_2023.dta", clear

* Estimate univariate linear regression
regress hhcon hhinc
```

```
      Source |       SS           df       MS      Number of obs   =       200
-------------+----------------------------------   F(1, 198)       =     57.08
       Model |  4842359.39         1  4842359.39   Prob > F        =    0.0000
    Residual |  16798378.5       198  84840.2954   R-squared       =    0.2238
-------------+----------------------------------   Adj R-squared   =    0.2198
       Total |  21640737.9       199  108747.427   Root MSE        =    291.27

------------------------------------------------------------------------------
       hhcon | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       hhinc |   .2705895   .0358165     7.55   0.000     .1999587    .3412203
       _cons |   268.0611   37.42847     7.16   0.000     194.2515    341.8707
------------------------------------------------------------------------------
```

R

```r
# Packages
library(haven)

# Load the Stata file
lcf_data <- read_dta("data/lcf_2023.dta")

# Estimate univariate linear regression
model1 <- lm(hhcon ~ hhinc, data = lcf_data)

# View results
summary(model1)
```

```
Call:
lm(formula = hhcon ~ hhinc, data = lcf_data)

Residuals:
    Min      1Q  Median      3Q     Max 
-519.70 -177.46  -60.08   95.78 1473.71 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 268.06105   37.42847   7.162 1.53e-11 ***
hhinc         0.27059    0.03582   7.555 1.52e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 291.3 on 198 degrees of freedom
Multiple R-squared:  0.2238,	Adjusted R-squared:  0.2198 
F-statistic: 57.08 on 1 and 198 DF,  p-value: 1.515e-12
```

Critical values

For normally distributed test-statistic, the critical value(s) for a two-sided $\alpha = 0.05$ (significance level) test are $\{-1.96, 1.96\}$.

The t-distribution is similarly symmetric:

```r
qt(0.025, df = 50)
```

```
[1] -2.008559
```

```r
qt(0.975, df = 50)
```

```
[1] 2.008559
```

And as $n - k - 1$ increases, the values converge to $\mathrm{z}_{0.975} = 1.96$. (More on this in Week 5.)

```r
qt(0.975, df = 100)
```

```
[1] 1.983972
```

```r
qt(0.975, df = 200)
```

```
[1] 1.971896
```

```r
qt(0.975, df = 400)
```

```
[1] 1.965912
```

```r
qnorm(0.975)
```

```
[1] 1.959964
```

Hypothesis tests using t-distribution are more 'conservative' relative to normal.

Useful code

Stata

```stata
* Standard normal distribution

** Evaluate point on CDF: i.e., probability below 0.675
dis normal(0.675)

** Percentile of CDF
dis invnormal(0.975)
```

```
* t-distribution

** Evaluate point on CDF (df = 100): i.e., probability below 0.675
dis t(100, 0.675)

** Percentile of CDF (df = 100)
dis invt(100, 0.975)
```

```
.75016212

1.959964

.74938326

1.9839715
```

R

```
# Standard normal distribution

## Evaluate point on CDF: i.e., probability below 0.675
pnorm(0.675)
```

```
[1] 0.7501621
```

```
## Percentile of CDF
qnorm(0.975)
```

```
[1] 1.959964
```

```
## t-distribution

## Evaluate point on CDF (df = 100): i.e., probability below 0.675
pt(0.675, 100)
```

```
[1] 0.7493833
```

```r
## Percentile of CDF (df = 100)
qt(0.975, 100)
```

```
[1] 1.983972
```